



PHYS121 Integrated Science-Physics

W4T1 Rotation

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
 - [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
 - [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)
- And others specified when needed.



9.4.3. An astronaut drops a golf ball that is initially at rest from a cliff on the surface of the moon. The ball falls freely under the influence of gravity. Which one of the following statements is true concerning the ball as it falls? Neglect any frictional effects.

- a) The ball will gain an equal amount of kinetic energy during each second.
- b) The ball will gain an equal amount of momentum during each second.
- c) The ball will gain an equal amount of momentum during each meter through which it falls.
- d) The ball will gain twice the amount of speed for each meter through which it falls.
- e) The amount of momentum the ball gains will be directly proportional to the amount of potential energy that it loses.

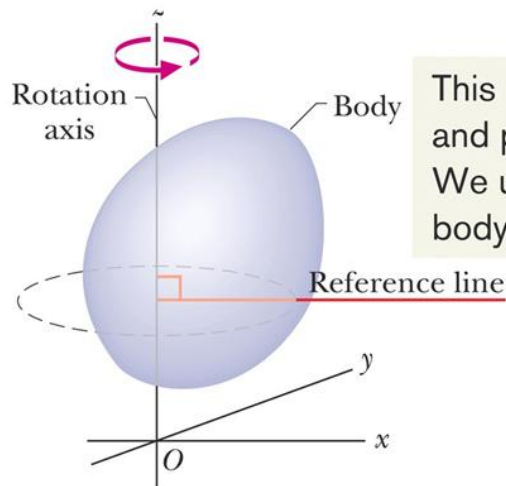


Learning Outcomes

- Describe rotation motions with comparison to translational motions
- Solve problems of constant angular acceleration.
- Calculate the rotational inertial and kinetic energy.
- Find the torque(s) and apply Newton's second law for rotation to solve problems

Rotational Variables

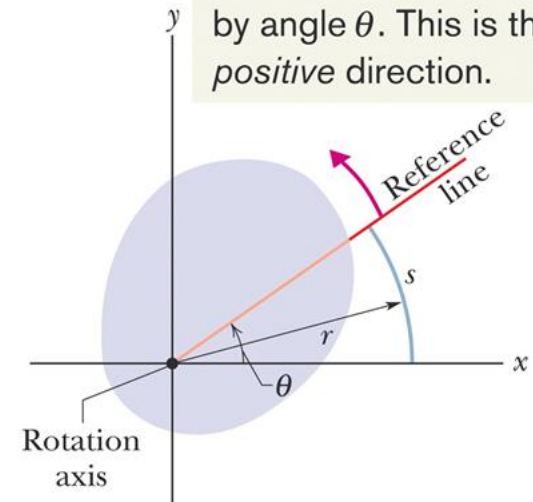
- The fixed axis is called the **axis of rotation**
- Figs 10-2, 10-3 show a reference line
- The **angular position** of this line (and of the object) is taken relative to a fixed direction, the **zero angular position**



This reference line is part of the body and perpendicular to the rotation axis. We use it to measure the rotation of the body relative to a fixed direction.

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Figure 10-2



The body has rotated *counterclockwise* by angle θ . This is the *positive* direction.

This dot means that the rotation axis is out toward you.

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Figure 10-3

- Measure ‘angle’ using **radians** (rad): dimensionless

$$\theta = \frac{s}{r} \quad (\text{radian measure}). \quad \text{Equation (10-1)}$$

$$1 \text{ rev} = 360^\circ = \frac{2\pi r}{r} = 2\pi \text{ rad}, \quad \text{Equation (10-2)}$$

- Do not reset θ to zero after a full rotation
- We know all there is to know about the kinematics of rotation if we have $\theta(t)$ for an object
- Define angular displacement as: $\Delta\theta = \theta_2 - \theta_1$.
- “Clocks are negative” (*chosen*): Equation (10-4)

An angular displacement in the counterclockwise direction is positive, and one in the clockwise direction is negative.

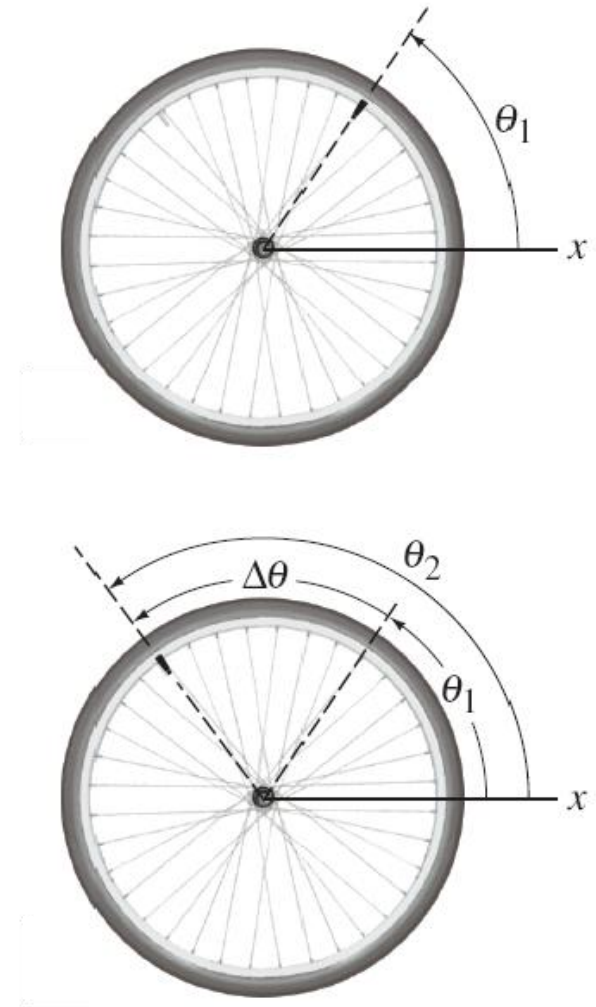
- **Average angular velocity:**
angular displacement during
a time interval

$$\omega_{\text{avg}} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t},$$

- **Instantaneous angular velocity:** limit as $\Delta t \rightarrow 0$

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}.$$

- *If the body is rigid, these equations hold
for all points on the body*
- Magnitude of angular velocity = **angular
speed**



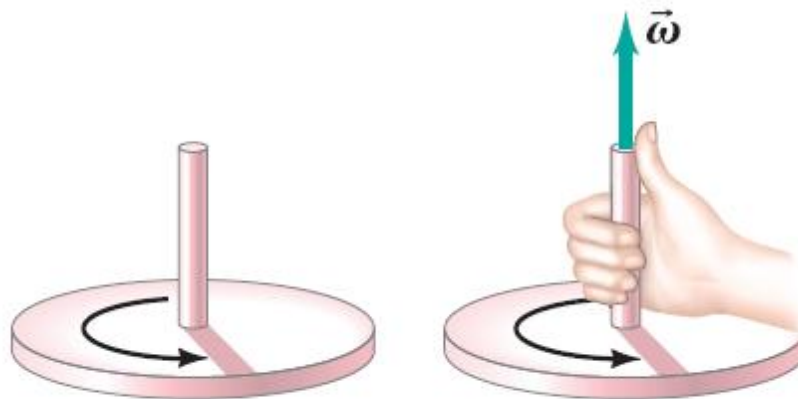
- **Average angular acceleration:** angular velocity change during a time interval

$$\alpha_{\text{avg}} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t}, \quad \text{Equation (10-7)}$$

- **Instantaneous angular acceleration :** limit as $\Delta t \rightarrow 0$

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}. \quad \text{Equation (10-8)}$$

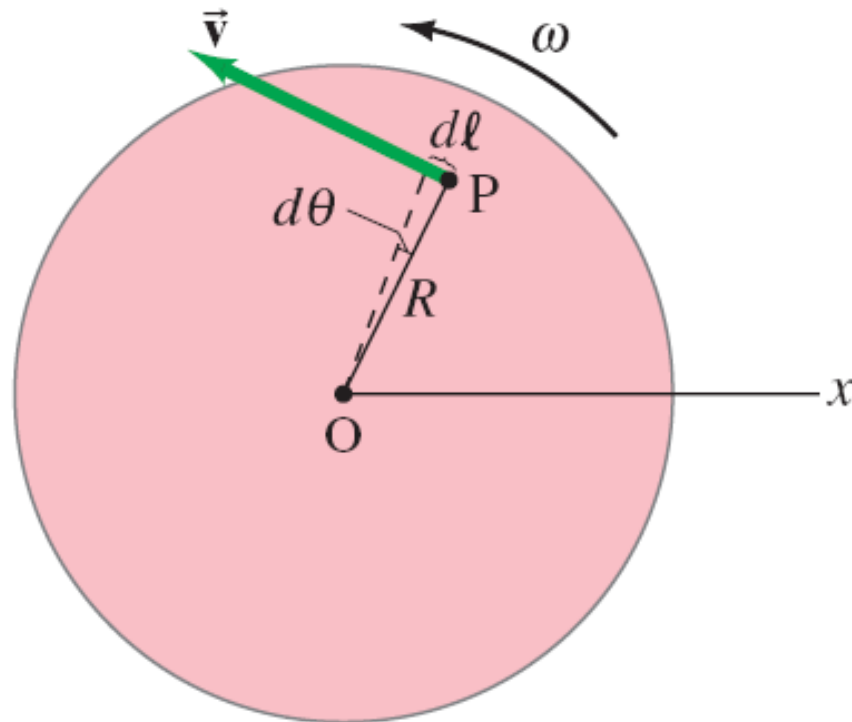
- If the body is rigid, these equations hold for all points on the body
- With right-hand rule to determine direction, angular velocity & acceleration can be written as vectors
- If the body rotates around the vector, then the vector points along the axis of rotation
- *Angular displacements are not vectors, because the order of rotation matters for rotations around different axes*

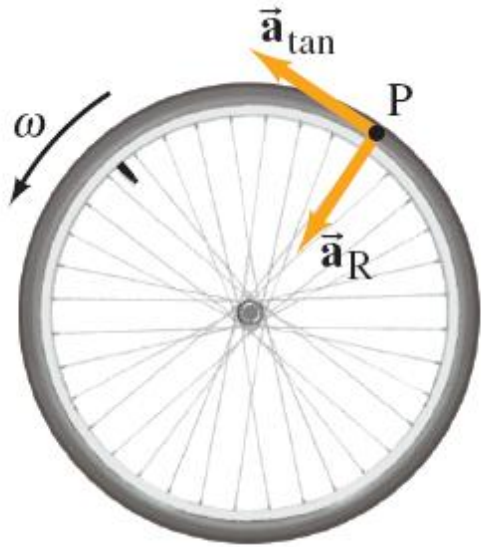


Every point on a rotating body has an angular velocity ω and a linear velocity v (with respective to the rotation center).

$$v = R\omega.$$

They are related:





If the angular velocity of a rotating object changes (so as the linear velocity), it has a tangential acceleration:

$$a_{\text{tan}} = \frac{dv}{dt} = R \frac{d\omega}{dt} = R\alpha.$$

Even if the angular velocity is constant, each point on the object has a centripetal acceleration:

$$a_R = \frac{v^2}{R} = \frac{(R\omega)^2}{R} = \omega^2 R.$$

Relating the Linear and Angular Variables

Checkpoint 3

A cockroach rides the rim of a rotating merry-go-round. If the angular speed of this system (merry-go-round + cockroach) is constant, does the cockroach have (a) radial acceleration and (b) tangential acceleration? If ω is decreasing, does the cockroach have (c) radial acceleration and (d) tangential acceleration?

Answer:

- (a) yes
- (b) no
- (c) yes
- (d) yes

Here is the correspondence between linear and rotational quantities (case by case):

TABLE 10–1
Linear and Rotational Quantities

Linear	Type	Rotational	Relation (θ in radians)
x	displacement	θ	$x = R\theta$
v	velocity	ω	$v = R\omega$
a_{tan}	acceleration	α	$a_{\text{tan}} = R\alpha$

The frequency is the number of complete revolutions per second:

$$f = \frac{\omega}{2\pi}.$$

Frequencies are measured in hertz:

$$1 \text{ Hz} = 1 \text{ s}^{-1}.$$

The period is the time one revolution takes:

$$T = \frac{1}{f}.$$

Example: Given ω as function of time.

A disk of radius $R = 3.0$ m rotates at an angular velocity $\omega = (1.6 + 1.2t)$ rad/s, where t is in seconds. At the instant $t = 2.0$ s, determine (a) the angular acceleration, and (b) the speed v and the components of the acceleration a of a point on the edge of the disk.

Rotation with Constant Angular Acceleration

Table 10-1 Equations of Motion for Constant Linear Acceleration and for Constant Angular Acceleration

Equation Number	Linear Equation	Missing Variable	Missing Variable	Angular Equation	Equation Number
(2-11)	$v = v_0 + at$	$x - x_0$	$\theta - \theta_0$	$\omega = \omega_0 + \alpha t$	(10-12)
(2-15)	$x - x_0 = v_0 t + \frac{1}{2} at^2$	v	ω	$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$	(10-13)
(2-16)	$v^2 = v_0^2 + 2a(x - x_0)$	t	t	$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$	(10-14)
(2-17)	$x - x_0 = \frac{1}{2}(v_0 + v)t$	a	α	$\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$	(10-15)
(2-18)	$x - x_0 = vt - \frac{1}{2} at^2$	v_0	ω_0	$\theta - \theta_0 = \omega t - \frac{1}{2} \alpha t^2$	(10-16)

Checkpoint 2

In four situations, a rotating body has angular position $\theta(t)$ given by (a) $\theta = 3t - 4$, (b) $\theta = -5t^3 + 4t^2 + 6$, (c) $\theta = \frac{2}{t^2} - \frac{4}{t}$, and (d) $\theta = 5t^2 - 3$. To which situations do the angular equations of Table 10-1 apply?

Answer:

Situations (a) and (d); the others do not have constant angular acceleration

Example: Centrifuge acceleration.

A centrifuge rotor is accelerated from rest to 20,000 rpm in 30 s. (a) What is its average angular acceleration? (b) Through how many revolutions has the centrifuge rotor turned during its acceleration period, assuming constant angular acceleration?

Kinetic Energy of Rotation

- Apply the kinetic energy formula for a point particle and sum over all the particles

$$K = \sum \frac{1}{2} m_i v_i^2$$

- different linear velocities (same angular velocity for all particles but possibly different radii)
- Then write velocity in terms of angular velocity:

$$K = \sum \frac{1}{2} m_i (\omega r_i)^2 = \frac{1}{2} \left(\sum m_i r_i^2 \right) \omega^2, \quad \text{Equation (10-32)}$$

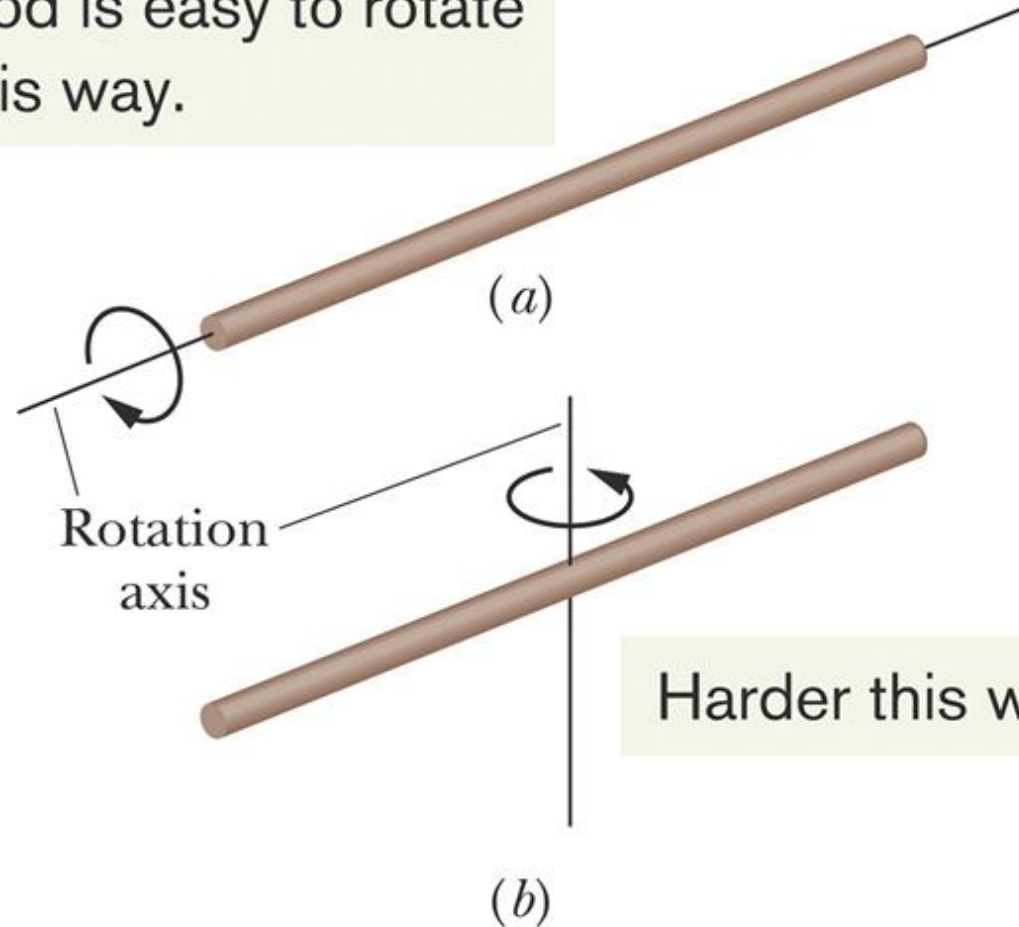
We call the quantity in parentheses on the right side the **rotational inertia**, or **moment of inertia**, I

- *This is a constant for a rigid object and given rotational axis*
- Caution: the axis for I must always be specified
- We can write: $I = \sum m_i r_i^2$ (rotational inertia)
- And rewrite the kinetic energy as:

$$K = \frac{1}{2} I \omega^2 \quad (\text{radian measure})$$

- Use these equations for a finite set of rotating particles
- Rotational inertia corresponds to how difficult it is to change the state of rotation (speed up, slow down or change the axis of rotation)

Rod is easy to rotate
this way.



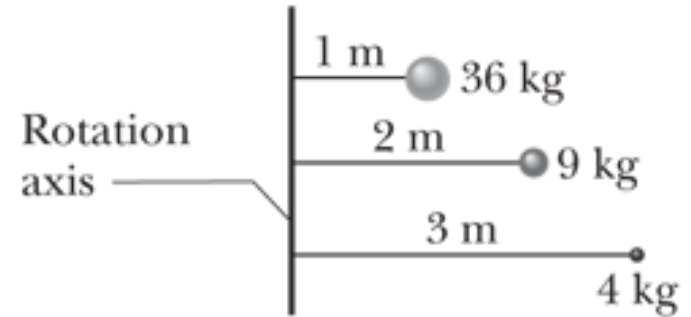
Harder this way.

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Figure 10-11

Checkpoint 4

The figure shows three *small* spheres that rotate about a vertical axis. The perpendicular distance between the axis and the center of each sphere is given. Rank the three spheres according to their rotational inertia about that axis, greatest first.



Answer:

They are all equal!

- Integrating Equation.10-33 over a continuous body:

$$I = \int r^2 dm \quad (\text{rotational inertia, continuous body}). \quad \text{Equation (10-35)}$$

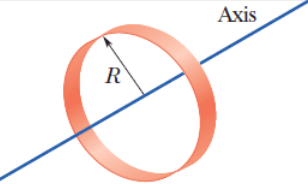
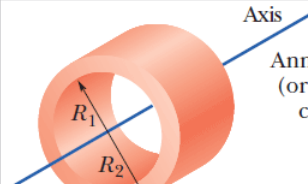
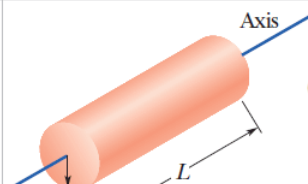
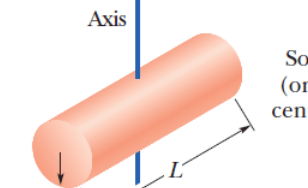
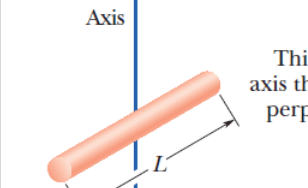
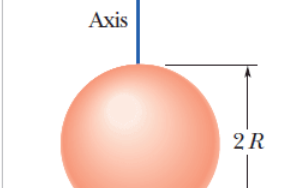
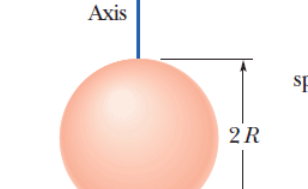
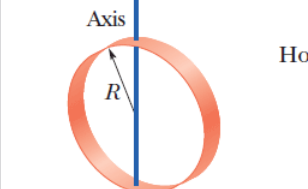
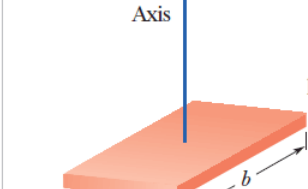
 Hoop about central axis $I = MR^2$ (a)	 Annular cylinder (or ring) about central axis $I = \frac{1}{2}M(R_1^2 + R_2^2)$ (b)	 Solid cylinder (or disk) about central axis $I = \frac{1}{2}MR^2$ (c)
 Solid cylinder (or disk) about central diameter $I = \frac{1}{4}MR^2 + \frac{1}{12}ML^2$ (d)	 Thin rod about axis through center perpendicular to length $I = \frac{1}{12}ML^2$ (e)	 Solid sphere about any diameter $I = \frac{2}{5}MR^2$ (f)
 Thin spherical shell about any diameter $I = \frac{2}{3}MR^2$ (g)	 Hoop about any diameter $I = \frac{1}{2}MR^2$ (h)	 Slab about perpendicular axis through center $I = \frac{1}{12}M(a^2 + b^2)$ (i)

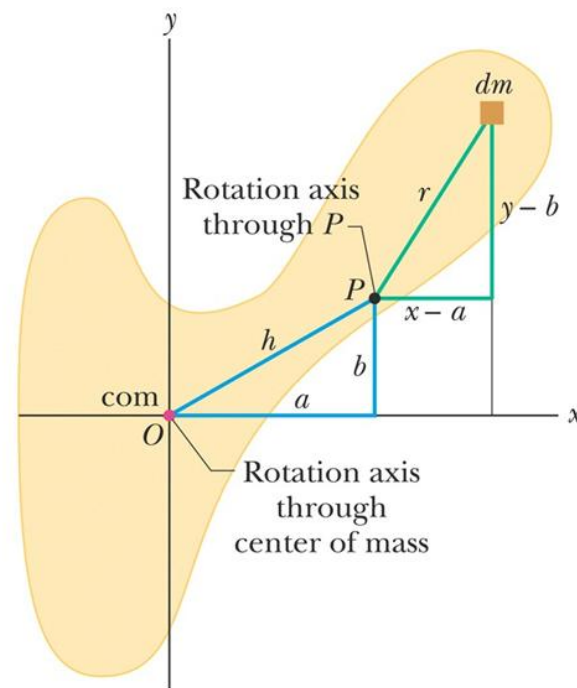
Table 10-2 Some Rotational Inertias

- If we know the moment of inertia for the center of mass axis, we can find the moment of inertia for a parallel axis with the **parallel-axis theorem**:

$$I = I_{\text{com}} + Mh^2 \quad \text{Equation (10-36)}$$

- Note the axes must be parallel, and the first must go through the center of mass
- This does not relate the moment of inertia for two arbitrary axes

We need to relate the rotational inertia around the axis at P to that around the axis at the com.

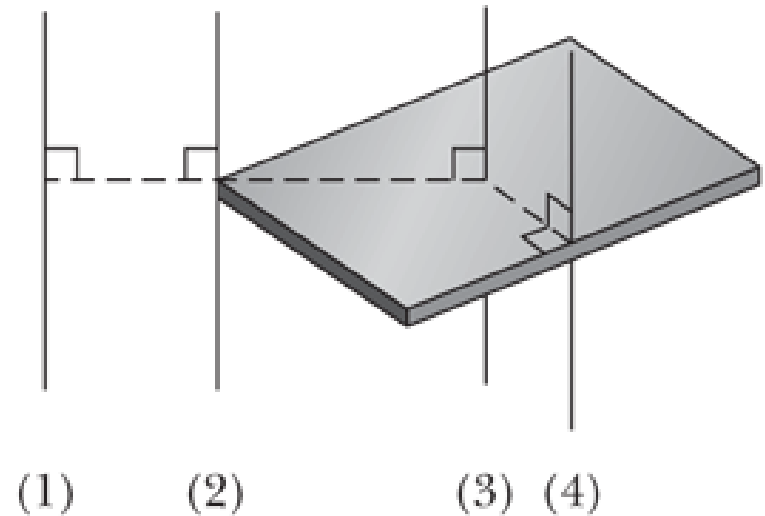


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Figure 10-12

Checkpoint 5

The figure shows a book-like object (one side is longer than the other) and four choices of rotation axes, all perpendicular to the face of the object. Rank the choices according to the rotational inertia of the object about the axis, greatest first.

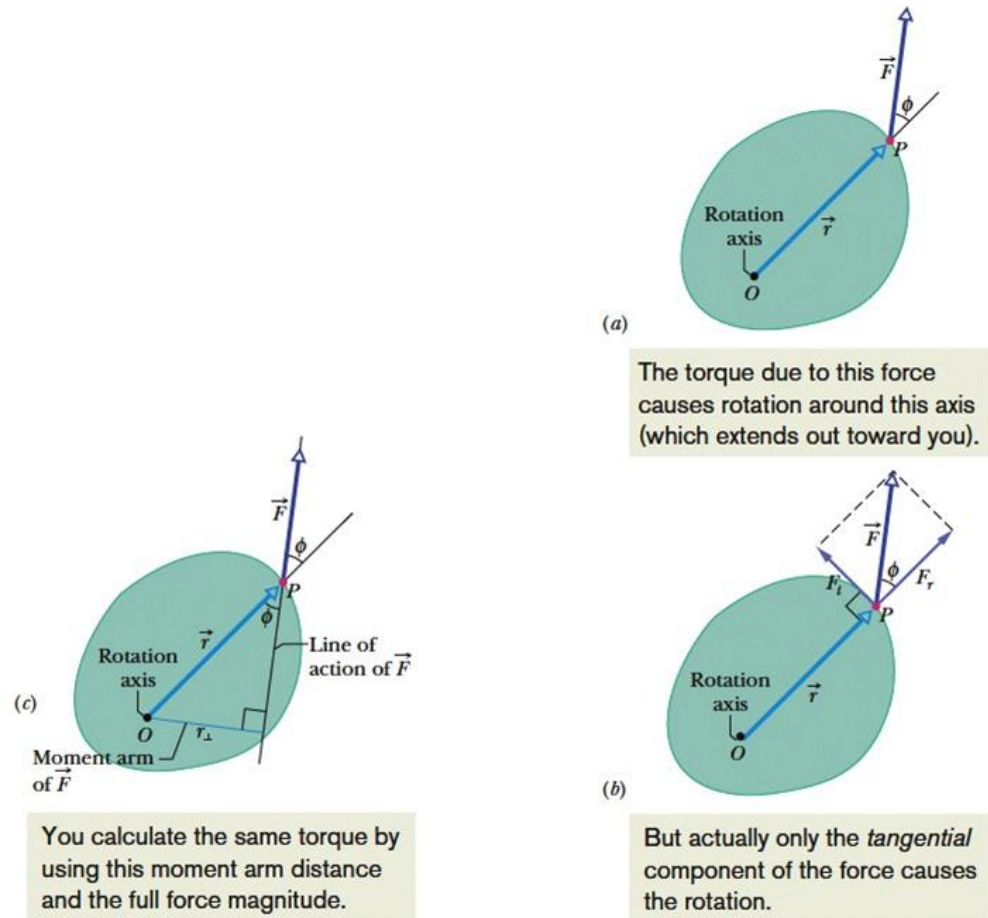


Answer:

(1), (2), (4), (3)

Torque

- The force necessary to rotate (change the rotation status) an object depends on the *angle* of the force and *where it is applied*
- We can resolve the force into components to see how it affects rotation



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Figure 10-16

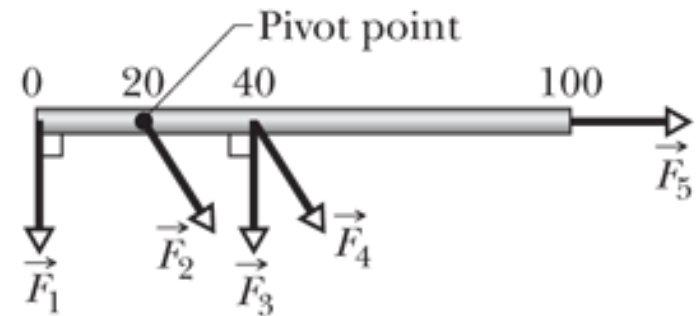
- **Torque** takes these factors into account:

$$\tau = (r)(F \sin \phi). \quad \text{Equation (10-39)}$$

- A line extended through the applied force is called the **line of action** of the force
- The perpendicular distance from the line of action to the axis is called the **moment arm (level arm)**
- The unit of torque is the newton-meter, N•m
- Note that 1 J = 1 N•m, but torques are never expressed in joules, torque is not energy
- Again, torque is positive if it would cause a counterclockwise rotation, otherwise negative
- For several torques, the **net torque** or **resultant torque** is the sum of individual torques

Checkpoint 6

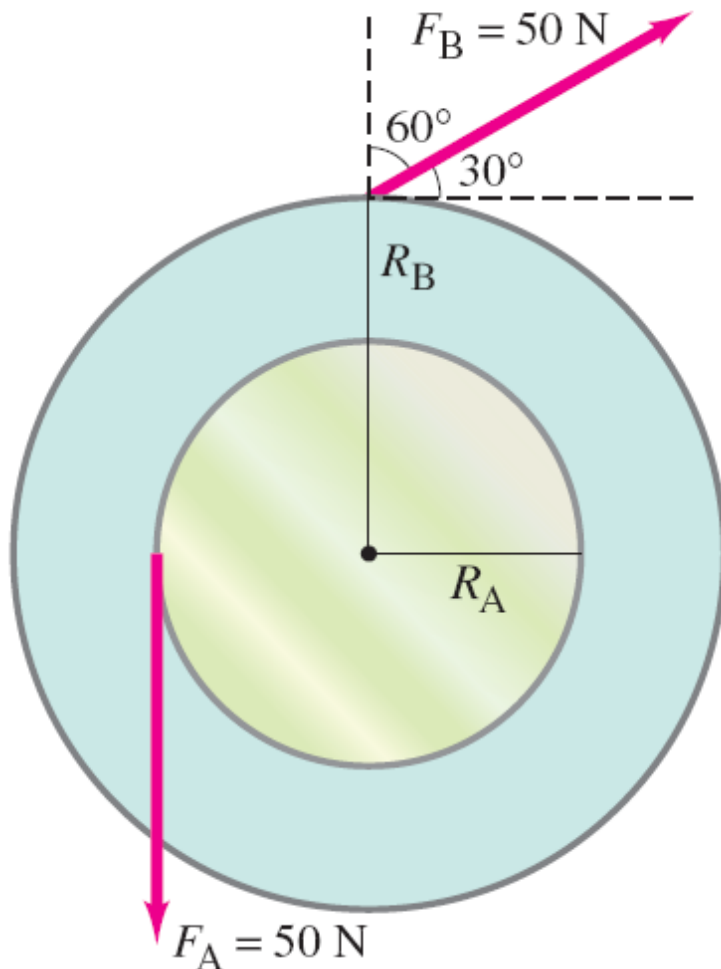
The figure shows an overhead view of a meter stick that can pivot about the dot at the position marked 20 (for 20 cm). All five forces on the stick are horizontal and have the same magnitude. Rank the forces according to the magnitude of the torque they produce, greatest first.



Answer:

F_1 & F_3 , F_4 , F_2 & F_5

Example: Torque on a compound wheel.



Two thin disk-shaped wheels, of radii $R_A = 30\text{ cm}$ and $R_B = 50\text{ cm}$, are attached to each other on an axle that passes through the center of each, as shown. Calculate the net torque on this compound wheel due to the two forces shown, each of magnitude 50 N .

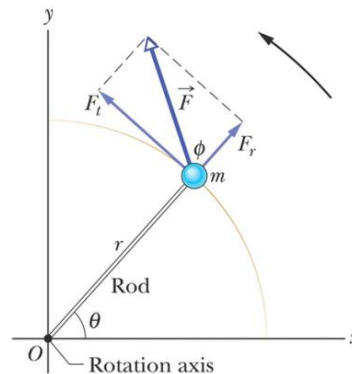
Newton's Second Law for Rotation

- Rewrite $F = ma$ with rotational variables:

$$\tau_{\text{net}} = I\alpha \quad \text{Equation (10-42)}$$

- *It is torque that causes angular acceleration*

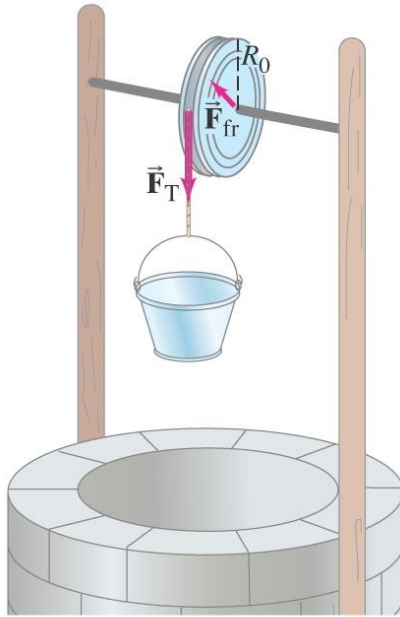
The torque due to the tangential component of the force causes an angular acceleration around the rotation axis.



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Figure 10-17

Example: Pulley and bucket.



(a)



(b)

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A bucket of weight $w=15.0\text{ N}$ ($m=1.53\text{ kg}$) hanging from the cord. We assume the cord has negligible mass and does not stretch or slip on the pulley ($M=4.00\text{ kg}$, $R_0=33.0\text{ cm}$). (a) Calculate the angular acceleration of the pulley and the linear acceleration of the bucket. (b) Determine the angular velocity of the pulley and the linear velocity of the bucket at $t=3.00\text{ s}$ if the pulley (and bucket) start from rest at $t=0$. $\tau_{\text{friction}}=1.10\text{ m}\cdot\text{N}$ at the axle, $I_p=0.385\text{ kg}\cdot\text{m}^2$.



Learning Outcomes

- ✓ Describe rotation motions with comparison to translational motions)
- ✓ Solve problems of constant angular acceleration.
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- ✓ Find the torque(s) and apply Newton's second law for rotation to solve problems

CHAPTER REVIEW AND EXAMPLES, DEMOS

<https://openstax.org/books/university-physics-volume-1/pages/10-summary>

Since the direction of a positive angle in a circle is counterclockwise, we take counterclockwise rotations as being positive and clockwise rotations as negative.

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a}_t = \vec{\alpha} \times \vec{r}$$

Torque

When a force $\vec{F}$ is applied to a point P whose position is $\vec{r}$ relative to O (**Figure 10.32**), the torque $\vec{\tau}$ around O is

$$\vec{\tau} = \vec{r} \times \vec{F}. \quad (10.22)$$

Work-Energy Theorem for Rotation

The work-energy theorem for a rigid body rotating around a fixed axis is

$$W_{AB} = K_B - K_A \quad (10.29)$$

where

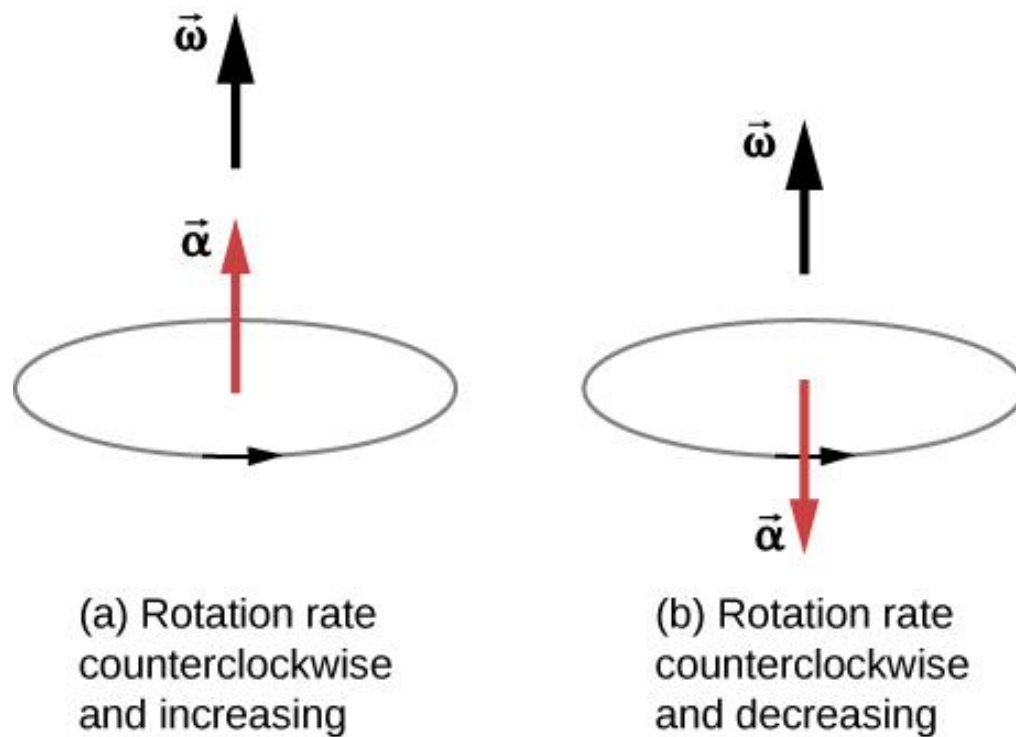
$$K = \frac{1}{2}I\omega^2$$

$$P = \tau\omega.$$

and the rotational work done by a net force rotating a body from point A to point B is

$$W_{AB} = \int_{\theta_A}^{\theta_B} \left(\sum_i \tau_i \right) d\theta. \quad (10.30)$$

FIGURE 10.7



The rotation is counterclockwise in both (a) and (b) with the angular velocity in the same direction.

- (a) The angular acceleration is in the same direction as the angular velocity, which increases the rotation rate.
- (b) The angular acceleration is in the opposite direction to the angular velocity, which decreases the rotation rate.

Example 10.10

Energy in a Boomerang

A person hurls a boomerang into the air with a velocity of 30.0 m/s at an angle of 40.0° with respect to the horizontal (**Figure 10.22**). It has a mass of 1.0 kg and is rotating at 10.0 rev/s. The moment of inertia of the boomerang is given as $I = \frac{1}{12}mL^2$ where $L = 0.7$ m. (a) What is the total energy of the boomerang when it leaves the hand? (b) How high does the boomerang go from the elevation of the hand, neglecting air resistance?

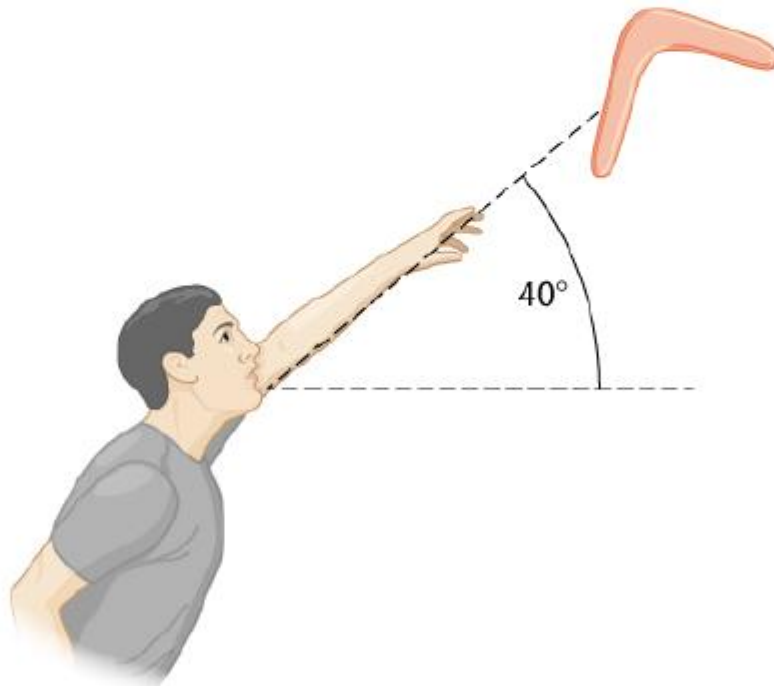
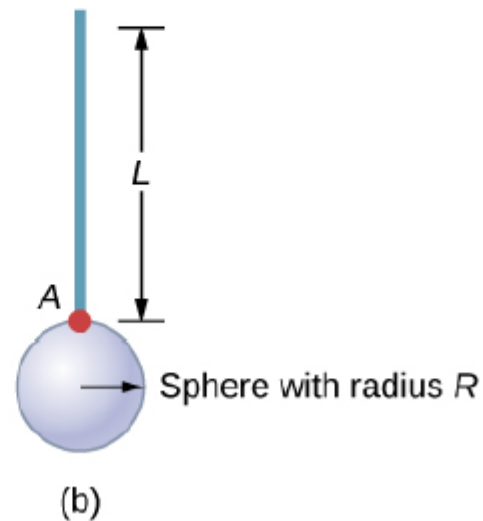
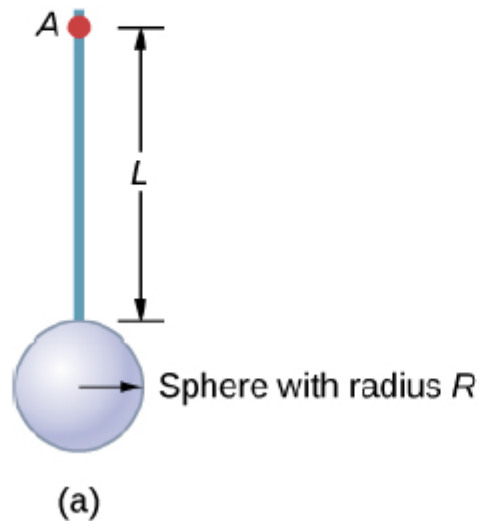


Figure 10.22 A boomerang is hurled into the air at an initial angle of 40° .

Example 10.12

Rod and Solid Sphere

Find the moment of inertia of the rod and solid sphere combination about the two axes as shown below. The rod has length 0.5 m and mass 2.0 kg. The radius of the sphere is 20.0 cm and has mass 1.0 kg.



Example 10.13

Angular Velocity of a Pendulum

A pendulum in the shape of a rod (**Figure 10.30**) is released from rest at an angle of 30° . It has a length 30 cm and mass 300 g. What is its angular velocity at its lowest point?

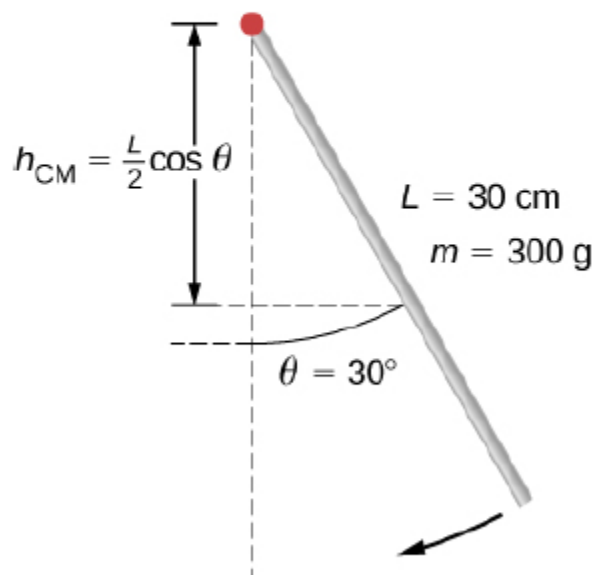


Figure 10.30 A pendulum in the form of a rod is released from rest at an angle of 30° .

Example 10.18

Rotational Work: A Pulley

A string wrapped around the pulley in **Figure 10.40** is pulled with a constant downward force $\vec{F}$ of magnitude 50 N. The radius R and moment of inertia I of the pulley are 0.10 m and $2.5 \times 10^{-3} \text{ kg}\cdot\text{m}^2$, respectively. If the string does not slip, what is the angular velocity of the pulley after 1.0 m of string has unwound? Assume the pulley starts from rest.

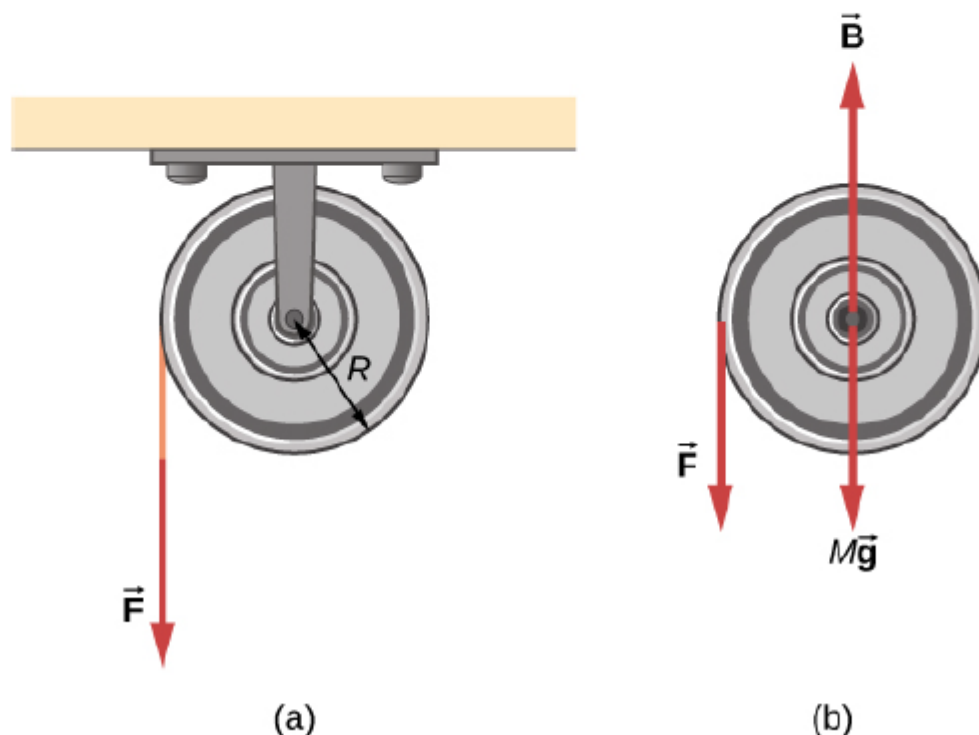


Figure 10.40 (a) A string is wrapped around a pulley of radius R .
(b) The free-body diagram.

Questions

